

FL-02

Prompt Iteration Log

A record of iterative prompt engineering applied to README generation

Task: README documentation writing for a portfolio website

Goal: Create a professional GitHub README file that explains the portfolio website, features, technology stack, installation steps, and usage clearly for developers and recruiters.

Overview

This log documents seven successive iterations of a single prompt engineering exercise. Each version applies one additional prompt engineering technique on top of the previous version, progressing from a naive, unguided prompt to a fully combined, structured prompt. For every iteration, the exact prompt, a condensed version of the model's output, and an observation on the resulting improvement are recorded.

- Naive Prompt — no technique applied
- Role Assignment — asking the AI to adopt an expert persona
- Context and Motivation — supplying background and purpose
- Few-Shot Examples — guiding style through implicit structure
- Output Structure — specifying an exact section order
- Step Decomposition — breaking the task into sequential steps
- Combined Techniques — merging role, context, requirements, structure, and step decomposition

Prompt Iterations

Version 0: Naive Prompt

TECHNIQUE: None (Before Prompt Engineering)

Prompt

Write a README file for my portfolio website.

Output (Condensed)

```
# Leon Infant B - Portfolio Website
Welcome to the repository for my personal portfolio website! This site
serves as a central hub to showcase my skills, experience, and the
various projects I've built as a Full-Stack Developer and B.E. Computer
Science student.

## Tech Stack & Skills
- Frontend: React.js, HTML5, CSS3, Modern UI/UX Design
- Backend: Node.js, Java
- Database: MongoDB

## Running the Portfolio Locally
1. git clone https://github.com/yourusername/portfolio.git
2. cd portfolio
3. npm install
4. npm start

## Contact & Connect
- GitHub: @yourusername
- LinkedIn: Leon Infant B
```

Observation

The AI created a basic README structure, but it used placeholder information like "yourusername" and "yourprofile". It did not know my actual portfolio details, technologies, projects, target audience, or preferred README format. The output was generic and needed more context.

Version 1: Role Assignment

TECHNIQUE: Role Assignment

Prompt

Act as a senior frontend developer and technical documentation expert.
Write a README file for my portfolio website.

Output (Condensed)

```
# Leon Infant B - Portfolio Website
Welcome to the repository for my personal portfolio website! This site
serves as a central hub to showcase my skills, experience, and the
various projects I've built as a Full-Stack Developer and B.E. Computer
Science student.
```

```
## About Me
I am a passionate full-stack developer with a strong foundation in
both front-end and back-end technologies.

## Tech Stack & Skills
- Frontend: React.js, HTML5, CSS3, Modern UI/UX Design
- Backend: Node.js, Java
- Database: MongoDB
- Concepts & Tools: OOP, System Design, UML Modeling

## Featured Projects
- LeoPizza, EECShop, Smart Waste Collection Notifier,
  Movie Ticket Booking Web, College Event Registration System

## Running the Portfolio Locally
1. git clone https://github.com/yourusername/portfolio.git
```

Observation

The response improved compared to Version 0 because the AI adopted the perspective of a senior frontend developer and created a more professional README structure, including sections like About Me, Tech Stack, and Featured Projects. However, it still made assumptions and did not know all my exact requirements because I did not provide detailed context.

Version 2: Context and Motivation

TECHNIQUE: Context and Motivation

Prompt

```
Act as a senior frontend developer and technical documentation expert.
I am a 2nd year B.E. Computer Science student building a personal
portfolio website to showcase my skills, projects, certifications, and
achievements for internship opportunities.
Create a professional GitHub README file for my portfolio website that
helps recruiters and developers understand my work.
```

Output (Condensed)

```
<h1 align="center">Hi there, I'm Leon Infant B</h1>
<h3 align="center">B.E. Computer Science Student | Aspiring Full-Stack Developer</h3>

## About Me
Currently a 2nd-year B.E. Computer Science student actively seeking
internship opportunities, passionate about full-stack web and mobile
application development.

## Tech Stack & Tools
- Frontend: React.js, HTML5, CSS3, Modern UI/UX Design
- Backend: Node.js, Java
- Databases: MongoDB
- Core Concepts: OOP, System Architecture, UML Modeling

## Featured Projects
```

```
LeoPizza | EECShop | Smart Waste Collection Notifier |  
Movie Ticket Booking Web | College Event Registration System
```

Certifications & Achievements

- [Certification Name] - [Issuing Organization] (Year)
- [Achievement/Hackathon Win] - [Short Description] (Year)

Local Development Setup

1. git clone .../portfolio.git
2. cd portfolio

Observation

The output became more personalized because I provided my background and the purpose of the README. The AI focused on recruiter-friendly content and included sections related to skills, projects, and achievements instead of creating a generic README.

Version 3: Few-Shot Examples

TECHNIQUE: Few-Shot Examples

Prompt

```
Act as a senior frontend developer and technical documentation expert.  
I am a 2nd year B.E. Computer Science student building a personal  
portfolio website for internship opportunities.  
Create a professional GitHub README file for my portfolio website.
```

Output (Condensed)

```
<h1 align="center">Leon Infant B – Personal Portfolio Website</h1>  
Navigation: About | Tech Stack | Projects | Features | Installation | Contact  
  
## Project Overview  
Built with clean architecture and modern UI/UX principles, serving as a  
central hub for recruiters and developers to explore my work.  
  
## About the Developer  
2nd-year B.E. Computer Science student actively seeking software  
engineering and full-stack development internships.  
  
## Technologies Used  
- Frontend: React.js, HTML5, CSS3  
- Backend: Node.js, Java  
- Database: MongoDB  
- Architecture & Design: OOP, System Architecture, UML Modeling  
  
## Key Projects  
1. LeoPizza 2. Movie Ticket Booking Web  
3. Smart Waste Collection Notifier 4. EECShop  
5. College Event Registration System  
  
## Features  
Responsive Design | Performance Optimized | Interactive UI |  
Project Showcase | Accessible
```

```
## Screenshots
(placeholder table for Home / Projects screenshots)

## Installation Steps
1. git clone .../portfolio.git
```

Observation

The output improved because examples guided the AI about the expected README structure and writing style. The AI included more relevant sections and created a more recruiter-focused document compared to previous versions.

Version 4: Output Structure

TECHNIQUE: Output Structure

Prompt

Act as a senior frontend developer and technical documentation expert.
I am a 2nd year B.E. Computer Science student creating a portfolio website for internship opportunities.
Create a professional GitHub README file for my portfolio website.

Follow this exact structure:

- | | |
|-----------------------------------|---------------------------|
| 1. Project Title and Introduction | 6. Featured Projects |
| 2. About Me | 7. Installation and Setup |
| 3. Features | 8. Usage Guide |
| 4. Technology Stack | 9. Future Improvements |
| 5. Project Screenshots | 10. Contact Information |

Write the README in a clean and professional style suitable for recruiters and developers.

Output (Condensed)

```
# Leon Infant B – Personal Portfolio Website
A live, interactive showcase of my technical skills, software
engineering projects, and professional journey as a Full-Stack Developer.

## About Me
2nd-year B.E. Computer Science student specializing in full-stack web
development and software architecture.

## Features
Fully Responsive Design | Performance Optimized | Interactive UI/UX |
Dynamic Project Showcase | Accessible Codebase

## Technology Stack
Frontend: React.js, HTML5, CSS3 | Backend: Node.js, Java |
Database: MongoDB | Methodologies: OOP, System Architecture, UML

## Project Screenshots
(placeholder table for Landing Page / Projects Directory)

## Featured Projects
```

```
LeoPizza, Movie Ticket Booking Web, Smart Waste Collection Notifier,  
EECSShop, College Event Registration System
```

```
## Installation and Setup Instructions  
1. git clone .../portfolio.git
```

Observation

The output became more organized because I specified the exact README structure. The AI followed the requested sections and created a document that was easier to read and use compared to previous versions.

Version 5: Step Decomposition

TECHNIQUE: Step Decomposition

Prompt

```
Act as a senior frontend developer and technical documentation expert.  
I am a 2nd year B.E. Computer Science student creating a portfolio  
website for internship opportunities.  
Help me create a professional GitHub README file.
```

Follow these steps:

```
Step 1: Analyze the purpose of the portfolio README and identify the  
        target audience.  
Step 2: List the important sections required for a recruiter-friendly  
        portfolio README.  
Step 3: Organize my portfolio details, skills, projects, and  
        technologies.  
Step 4: Create the complete README document using a professional format.  
Step 5: Review the README and suggest improvements.
```

```
Provide the final README after completing the analysis.
```

Output (Condensed)

```
# Leon Infant B – Portfolio Website Codebase  
Live demonstration of my capabilities in frontend design, component  
architecture, and clean code practices.  
> Live Demo: [Insert Link to Live Portfolio Here]  
  
## About Me  
2nd-year B.E. Computer Science student, aspiring Full-Stack Developer,  
seeking software engineering and web development internships.  
  
## Technical Arsenal  
Frontend: React.js, HTML5, CSS3 | Backend: Node.js, Java |  
Database: MongoDB | Core Concepts: OOP, System Design, UML  
  
## Featured Projects Showcase  
1. LeoPizza 2. EECSShop 3. Smart Waste Collection Notifier  
4. Movie Ticket Booking Web 5. College Event Registration System  
  
## Local Development Setup  
1. git clone .../portfolio.git
```

Observation

The output improved because the AI followed a structured problem-solving process. It first analyzed the requirements, identified important sections, and then created a more complete README. The response was more detailed and practical than previous versions.

Version 6: Final Combined Prompt

TECHNIQUE: Combined Prompt Engineering Techniques

Prompt

Act as a senior frontend developer and technical documentation expert.

Context: I am a 2nd year B.E. Computer Science student creating a personal portfolio website to showcase my skills, projects, certifications, and achievements for internship opportunities.

Goal: Create a professional GitHub README file that helps recruiters and developers understand my portfolio website.

Requirements:

- Include project introduction
- Explain my background
- Highlight technical skills
- Showcase important projects
- Mention technologies used
- Provide installation/usage steps
- Maintain a professional, developer-friendly style

Reference style: A good portfolio README should be clear, structured, visually organized, and focused on skills and achievements.

Follow this output structure:

1. Project Title and Introduction
2. About Me
3. Features
4. Technology Stack
5. Projects
6. Installation Steps
7. Usage Instructions
8. Future Improvements
9. Contact Information

Before writing the final README:

1. Analyze the requirements.
2. Identify the target audience.
3. Organize the required information.
4. Generate the final README.
5. Review it for improvements.

Output (Condensed)

```
# Leon Infant B – Personal Portfolio Website
A live, interactive showcase of my technical skills, software
engineering projects, and professional journey.
```

```
## About Me
```

2nd-year B.E. Computer Science student passionate about full-stack development and software architecture, actively seeking internships.

```
## Features
```

Responsive Design | Performance Focused | Interactive UI/UX |
Dynamic Project Showcase | Clean Codebase

```
## Technology Stack
Frontend: React.js, HTML5, CSS3 | Backend: Node.js, Java |
Database: MongoDB | Methodologies: OOP, System Architecture, UML
```

```
## Projects
LeoPizza, Movie Ticket Booking Web, Smart Waste Collection Notifier,
EECSShop, College Event Registration System
```

```
## Installation Steps
1. git clone .../portfolio.git
```

Observation

This final version produced the most complete and professional README. Combining role assignment, context, requirements, structure, and step-by-step reasoning resulted in a well-organized, recruiter-ready document with minimal placeholder content and a consistent, polished tone.

Cross-Model Comparison

Task: Generating a professional README for the portfolio website, compared across Claude (Anthropic) and ChatGPT (OpenAI).

Category	Claude (Anthropic)	ChatGPT (OpenAI)
Tone	Formal and documentation-focused; reads like professional developer documentation.	Conversational and recruiter-friendly; explains sections in a simpler, more approachable way.
Structure	Well-organized README with detailed explanations and smooth flow between sections.	Highly structured format with clear headings, bullet points, and easy-to-scan sections.
Accuracy	Followed the given portfolio context carefully and avoided adding unsupported details.	Generated useful suggestions quickly, but some sections required checking due to assumed details.
Strengths	Strong at producing polished documentation and maintaining consistency throughout.	Strong at quickly generating practical README structures and suggesting improvements.
Failure Points	Sometimes produced longer explanations than required.	Sometimes added generic information or assumptions that needed manual correction.

Final Reusable Prompt Template

Purpose: A reusable prompt for creating professional project README documentation.

Prompt

```
Act as a senior software developer and technical documentation expert.

Context:
I am creating documentation for a software project that needs to be
understood by developers, users, and recruiters.

Goal:
Create a professional GitHub README file that clearly explains the
project and improves its presentation.

Requirements:
- Include project overview          - Provide installation instructions
- Explain the problem the project solves - Explain how to use the application
- Describe key features              - Include screenshots/demo if available
- Mention technology stack          - Add future improvements
                                     - Provide contact/contribution info

Reference Style:
The README should be clear, structured, professional, and easy to
understand.

Output Format:
1. Project Title          6. Usage Instructions
2. Project Description    7. Screenshots/Demo
3. Features               8. Future Enhancements
4. Technology Stack       9. Contact Information
5. Installation Guide

Before writing:
1. Analyze the project purpose and target audience.
2. Identify important information required.
3. Organize the README structure.
4. Generate the final documentation.
5. Review the README and suggest improvements.
```

Sample Output — Applied to "Smart Waste Collection Notifier"

```
# Smart Waste Collection Notifier

## Project Description
Municipal waste management often suffers from inefficient routing,
leading to overflowing bins or unnecessary trips to empty containers.
This system monitors garbage bin capacities and automatically sends
real-time alerts when bins are full, optimizing collection routes and
reducing fuel consumption.

## Features
- Real-Time Capacity Alerts
- Interactive Dashboard for municipal managers
- Route Optimization Logic
- Responsive UI for desktop and mobile
```

Technology Stack

Frontend: React.js, HTML5, CSS3 | Backend: Node.js, Express.js

Database: MongoDB | Architecture: RESTful API, modular components

Installation Guide

1. `git clone .../smart-waste-notifier.git`
2. `cd smart-waste-notifier`

This template consolidates the most effective elements identified across all seven iterations — role assignment, contextual grounding, explicit requirements, a defined output structure, and step-by-step reasoning — into a single reusable prompt suitable for generating README documentation for any future project.